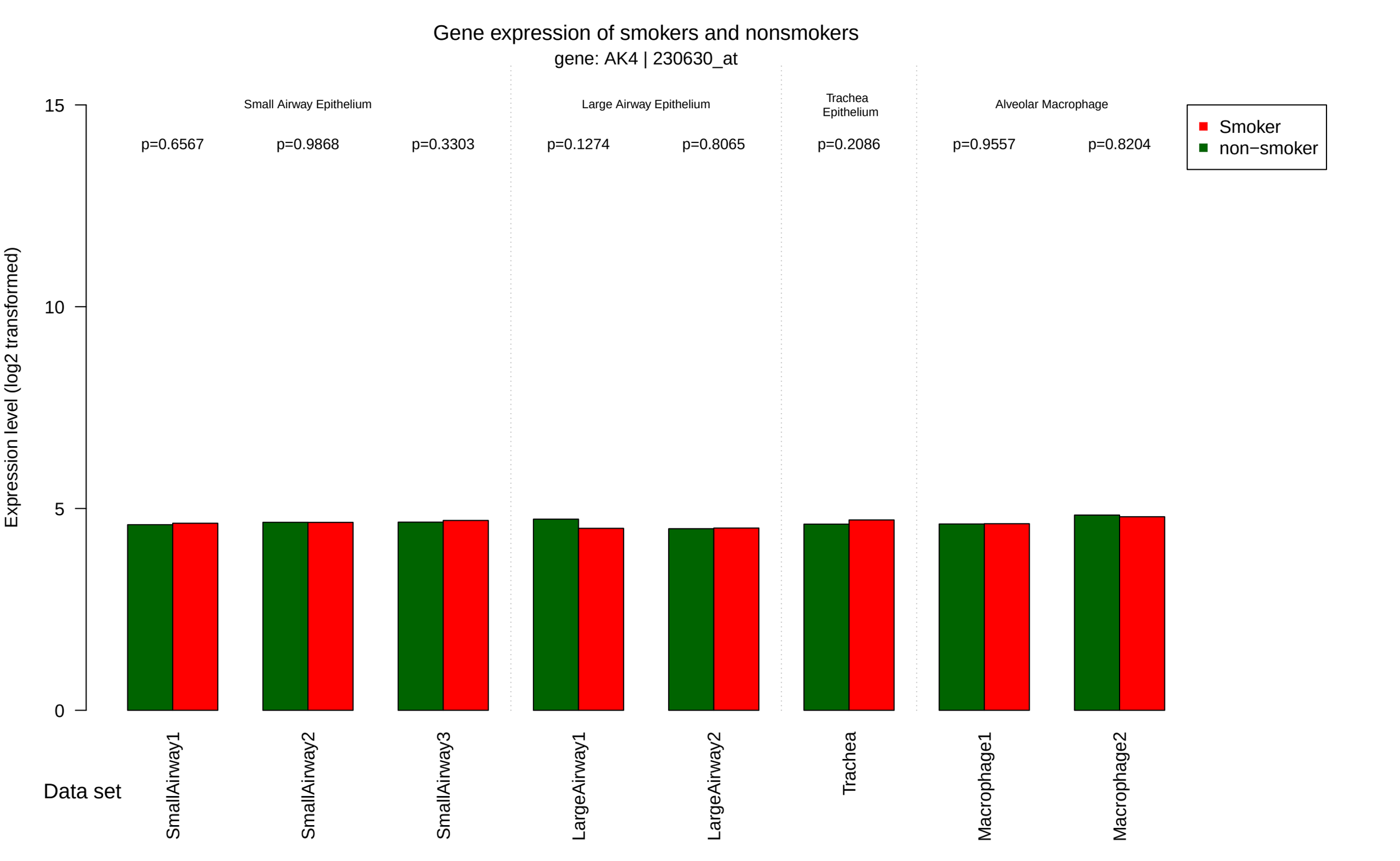
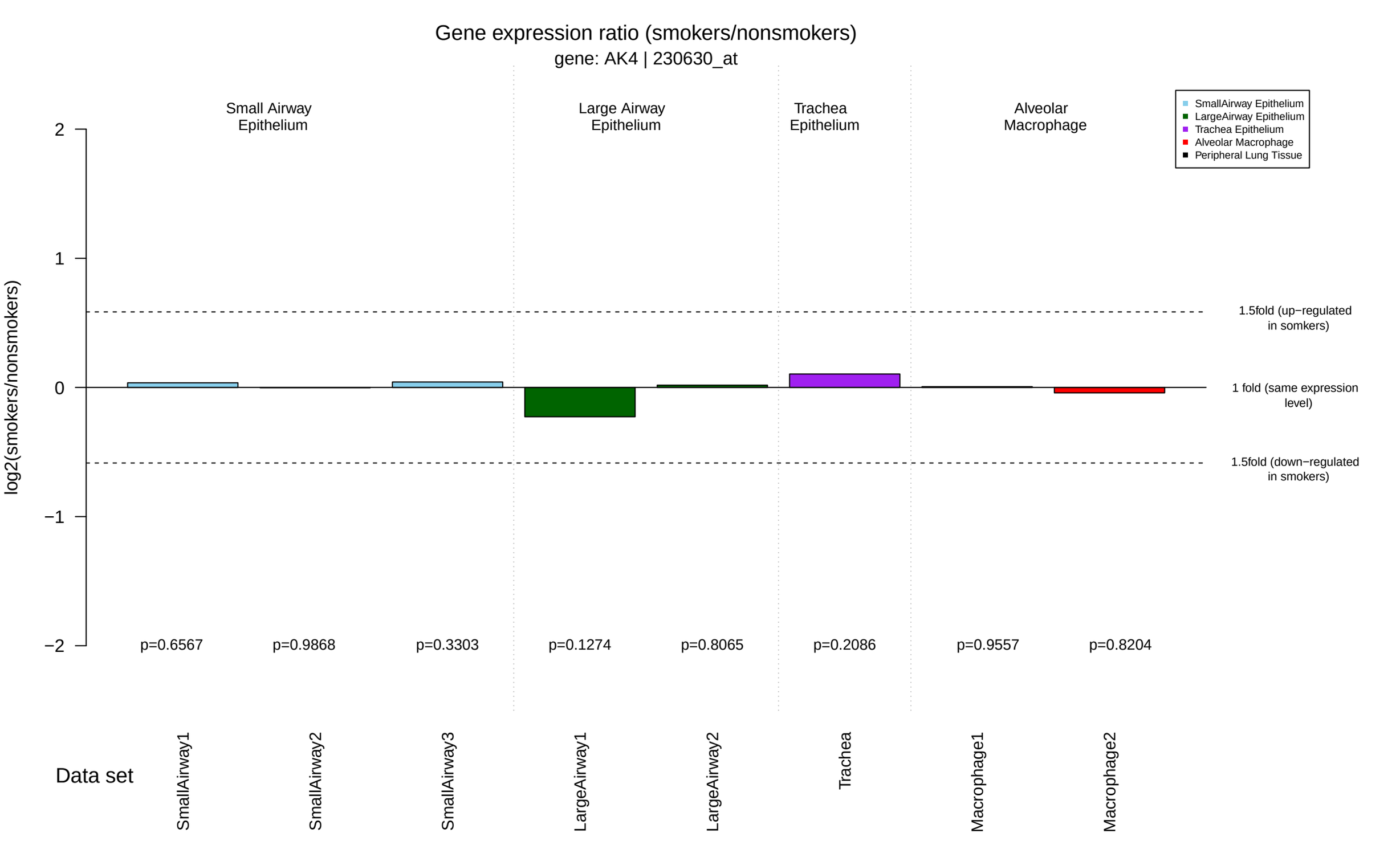
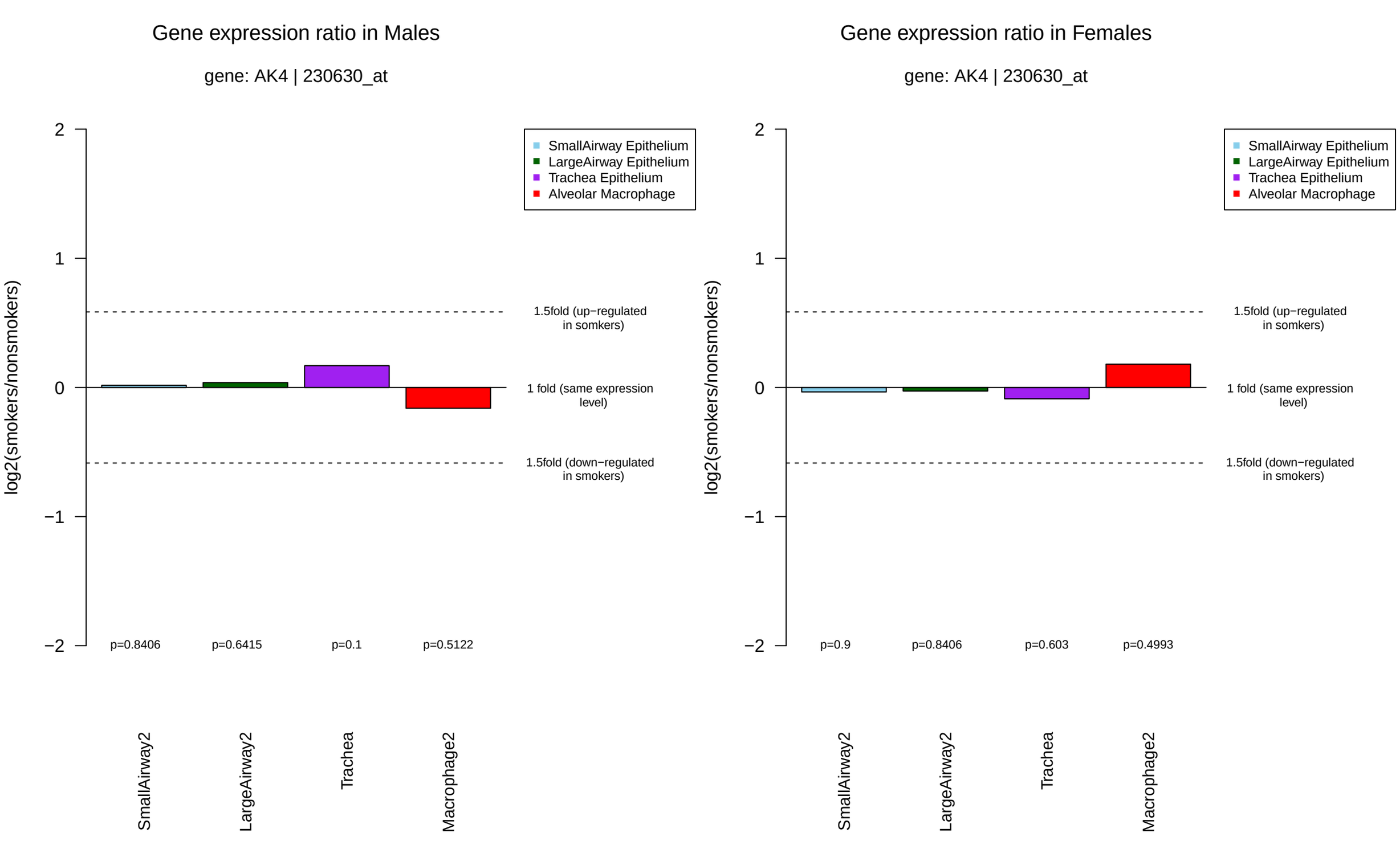
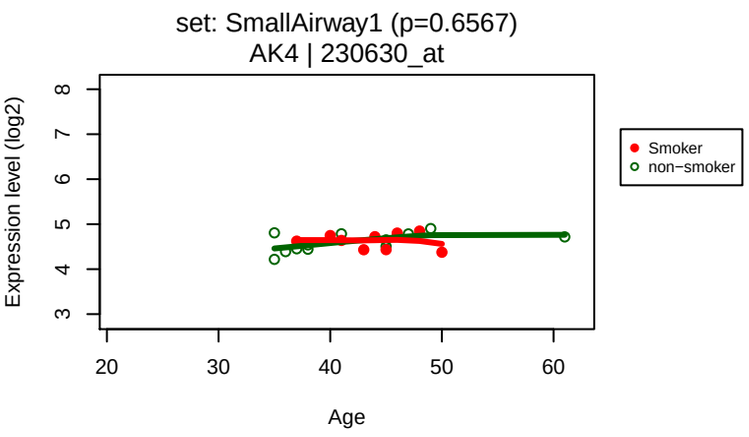
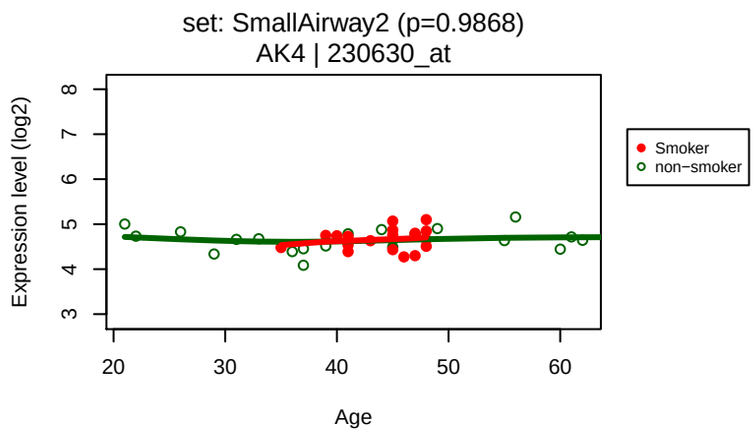
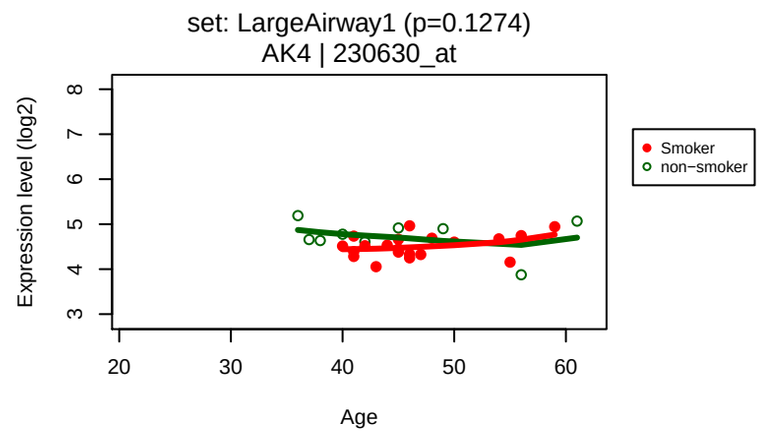

Gene expression of smokers and nonsmokers
gene: AK4 | 230630_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6567
p=0.9868
p=0.3303
p=0.1274
p=0.8065
p=0.2086
p=0.9557
p=0.8204
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

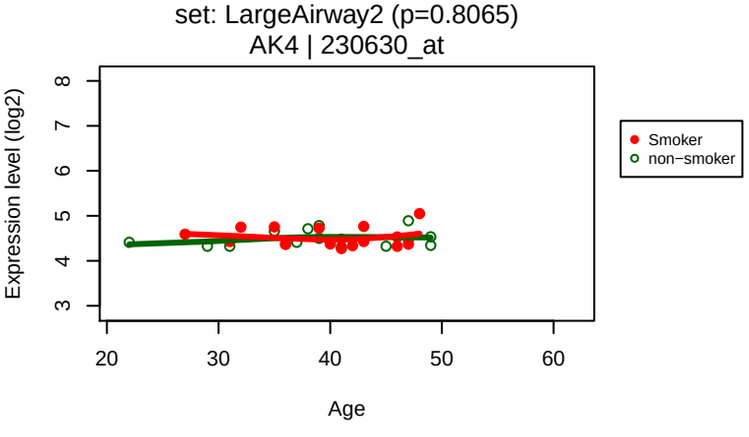

Gene expression ratio (smokers/nonsmokers)
gene: AK4 | 230630_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6567
p=0.9868
p=0.3303
p=0.1274
p=0.8065
p=0.2086
p=0.9557
p=0.8204
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

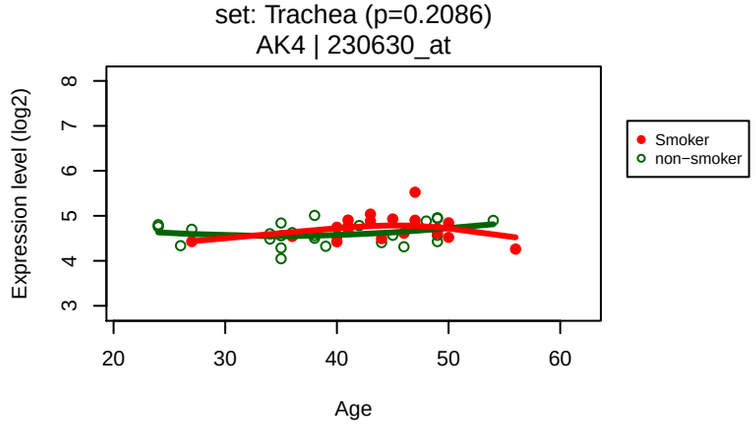

Gene expression ratio in Males
gene: AK4 | 230630_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8406
p=0.6415
p=0.1
p=0.5122
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK4 | 230630_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9
p=0.8406
p=0.603
p=0.4993
SmallAirway2
LargeAirway2
Trachea
Macrophage2

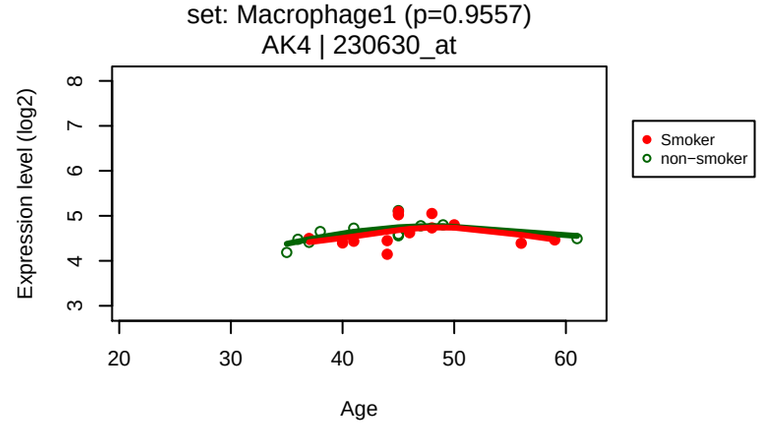

set: SmallAirway1 (p=0.6567)
AK4 | 230630_at
Expression level (log2)
Age
Smoker
non−smoker

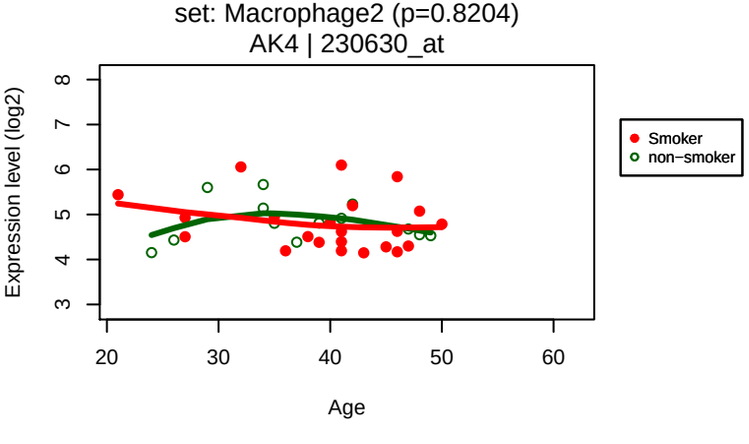

set: SmallAirway2 (p=0.9868)
AK4 | 230630_at
Expression level (log2)
Age
Smoker
non−smoker

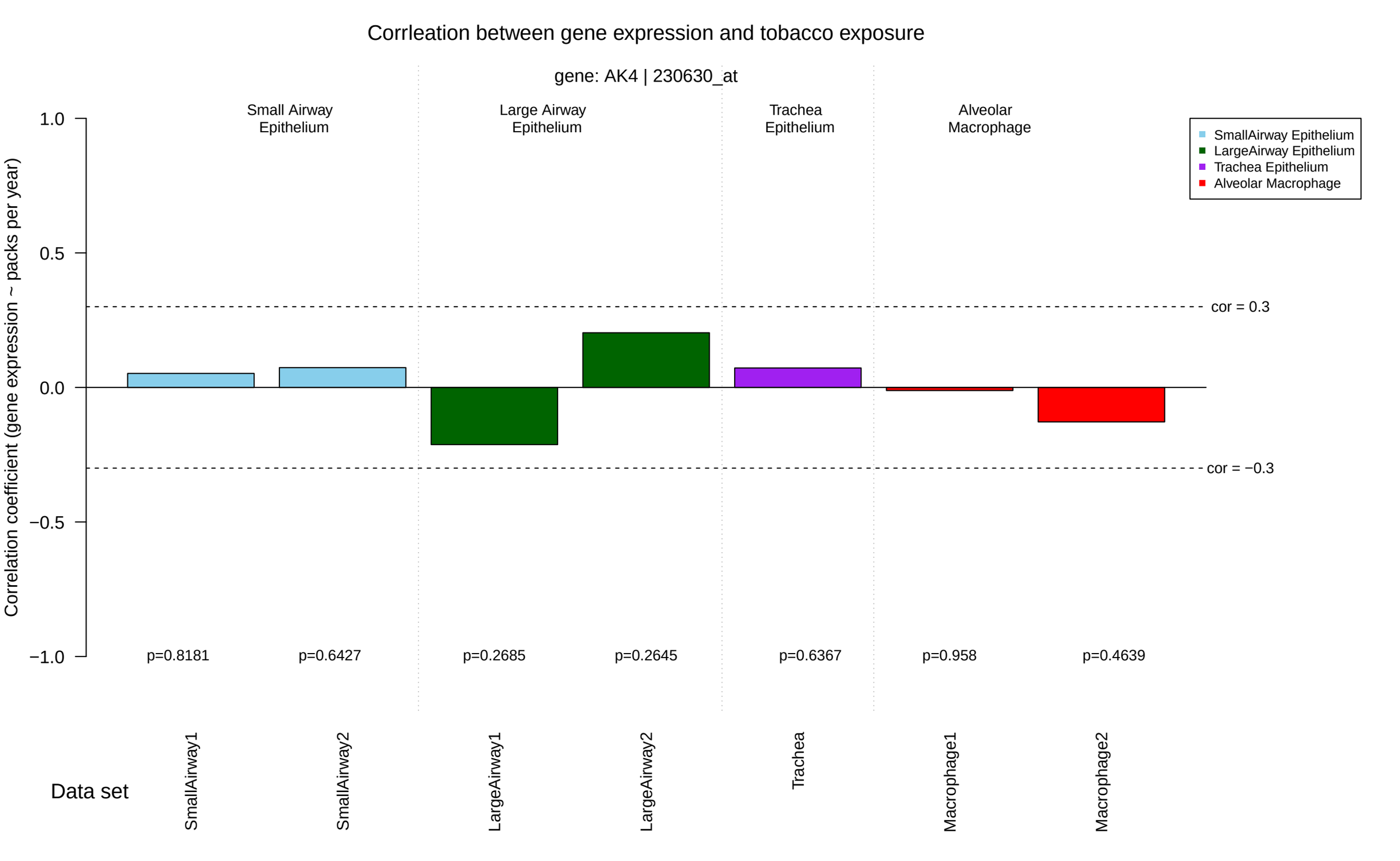

set: LargeAirway1 (p=0.1274)
AK4 | 230630_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8065)
AK4 | 230630_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2086)
AK4 | 230630_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9557)
AK4 | 230630_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8204)
AK4 | 230630_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK4 | 230630_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8181
p=0.6427
p=0.2685
p=0.2645
p=0.6367
p=0.958
p=0.4639
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2